RTL-to-GDSII Physical Design of a Serial-Parallel Multiplier (SPM) Using OpenLane and SkyWater 130nm PDK

Domain

VLSI Digital Design & ASIC Physical Implementation

Technology

SkyWater 130nm CMOS

Author

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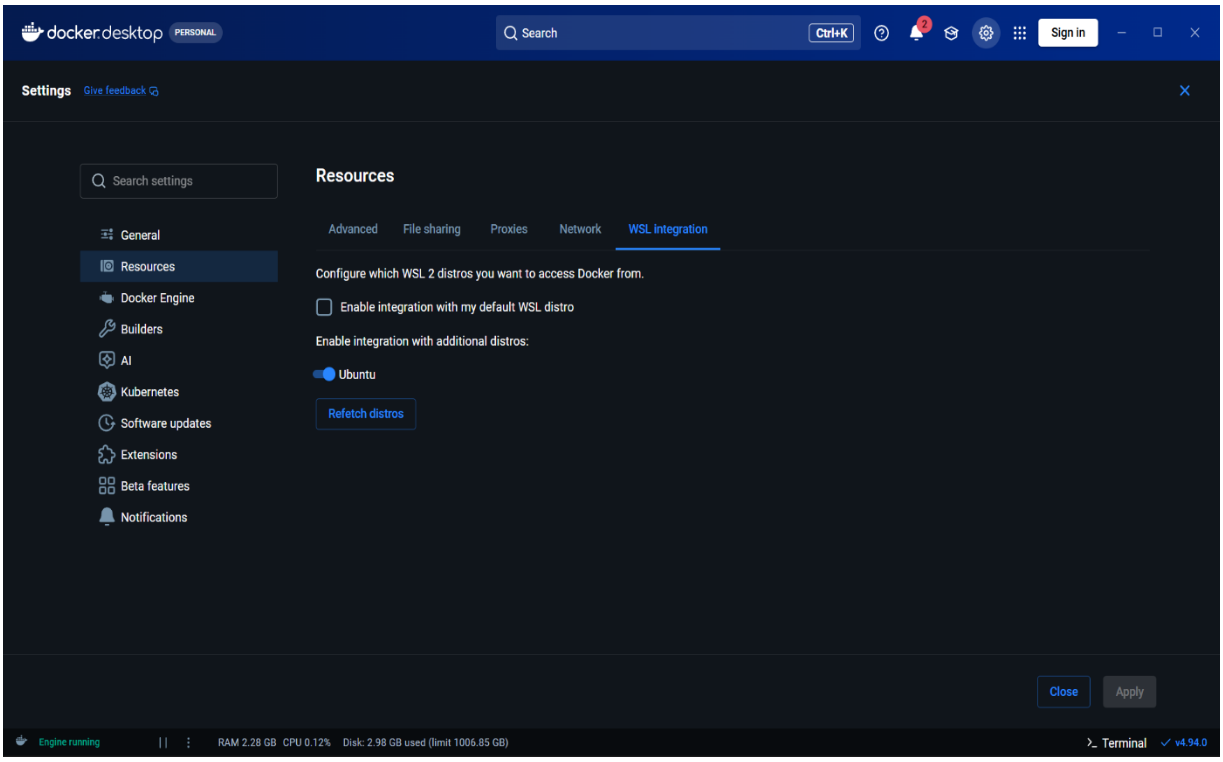
Abstract

This project demonstrates an automated RTL-to-GDSII physical implementation flow for an arithmetic Serial-Parallel Multiplier (SPM) architecture. The complete ASIC design workflow is executed using open-source EDA tools integrated under the OpenLane framework targeting the SkyWater 130nm Process Design Kit (PDK). The implementation covers logic synthesis, floorplanning, standard cell placement, clock tree synthesis (CTS), global and detailed routing, followed by physical sign-off verification. The resulting mask layout achieved zero setup and hold timing violations and zero Design Rule Check (DRC) violations, delivering a fabrication-ready GDSII file.

Toolchain & Experimental Setup

The implementation was executed on an open-source development stack running inside a containerized Linux environment on Windows 11:

- Host Operating System: Windows 11
- Linux Subsystem: Windows Subsystem for Linux 2 (WSL2 - Ubuntu Backend)
- Containerization: Docker Desktop integrated with WSL2 engine
- Synthesis Engine: Yosys (RTL synthesis) & ABC (technology mapping)
- Physical Implementation: OpenROAD (Floorplan, Placement, CTS, Routing)
- Design Rule Checking: Magic VLSI
- Layout Viewer: KLayout v0.30


docker desktop PERSONAL
Search
Ctrl+K
Sign in
Settings
Give feedback
Search settings
General
Resources
Docker Engine
Builders
AI
Kubernetes
Software updates
Extensions
Beta features
Notifications
Resources
Advanced
File sharing
Proxies
Network
WSL integration
Refetch distros
Close
Apply
Terminal
v4.94.0

Configure which WSL 2 distros you want to access Docker from.

☐ Enable integration with my default WSL distro

Enable integration with additional distros:

● Ubuntu

Engine running RAM 2.28 GB CPU 0.12% Disk: 2.98 GB used (limit 1006.85 GB)

Figure 1: Docker Desktop environment integrated with WSL2 Ubuntu backend

Architectural Overview of SPM

The Serial-Parallel Multiplier (SPM) is an arithmetic building block designed for area-efficient digital processing.

- Working Principle: One operand is loaded in parallel into internal registers, while the second operand is fed serially bit-by-bit per clock cycle.
- Hardware Utilization: Uses shift registers, full adders, and accumulation logic to compute the product iteratively.
- Application: Ideal for DSP applications, FIR filtering, and resource-constrained micro-architectures where full-parallel tree multipliers cause silicon area and wiring congestion penalties.

Physical Design Flow Execution

➢ Synthesis

The RTL code is synthesized using Yosys and mapped to SkyWater 130nm standard cells (sky130_fd_sc_hd). The gate-level netlist is verified for structural logic correctness.

➢ Floorplanning & PDN

Core utilization, aspect ratio, and die boundaries are configured. The Power Distribution Network (PDN) creates robust VDD and VSS rings and vertical/horizontal stripes to prevent IR drop.

➢ Placement

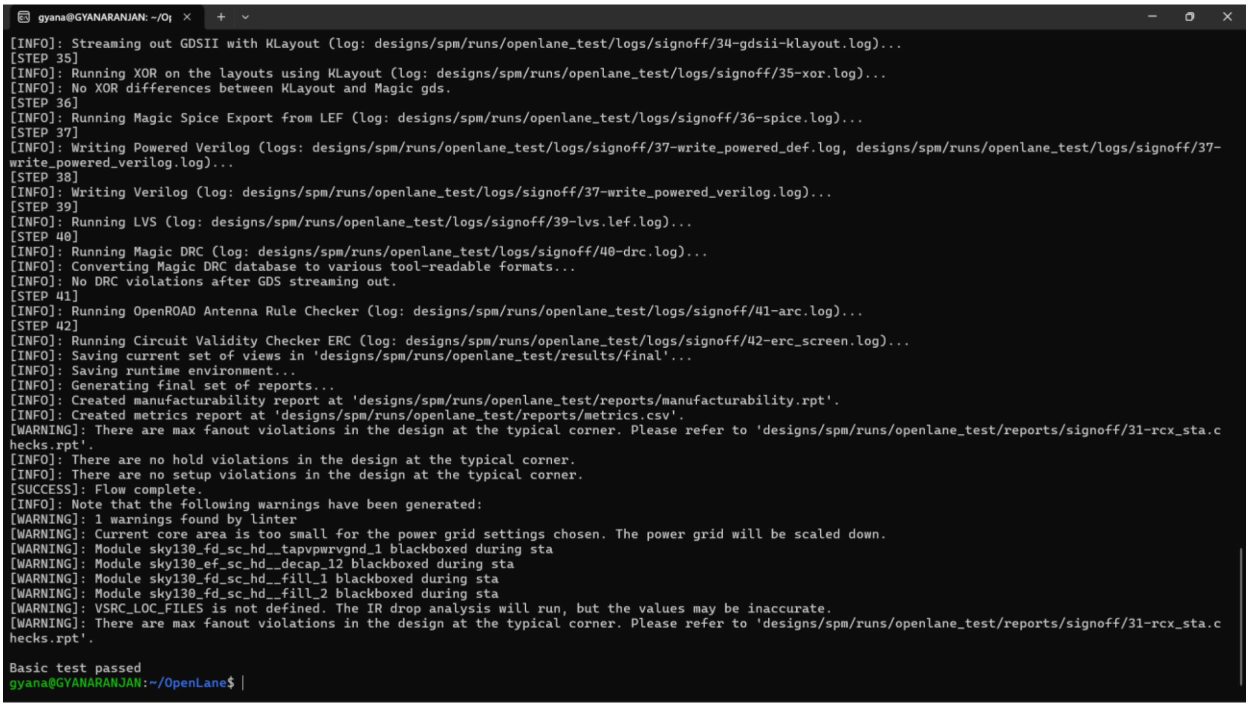
Global placement optimizes standard cell locations for wirelength, followed by detailed placement that aligns cells strictly within standard cell rows without overlaps.

➢ Clock Tree Synthesis (CTS)

TritonCTS constructs balanced clock trees using clock buffer inverters to ensure minimum clock skew and low insertion delay across all sequential flip-flops.

➢ Routing

TritonRoute performs global routing followed by detailed routing across multiple metal layers, satisfying design rules, minimum spacing, and via enclosure requirements.



```
[INFO]: Streaming out GDSII with KLayout (log: designs/spm/runs/openlane_test/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using KLayout (log: designs/spm/runs/openlane_test/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between KLayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/spm/runs/openlane_test/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/spm/runs/openlane_test/logs/signoff/37-write_powered_def.log, designs/spm/runs/openlane_test/logs/signoff/37-
write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/spm/runs/openlane_test/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/spm/runs/openlane_test/logs/signoff/39-lvs.lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/spm/runs/openlane_test/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/spm/runs/openlane_test/logs/signoff/41-arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/spm/runs/openlane_test/logs/signoff/42-erc_screen.log)...
[INFO]: Saving current set of views in 'designs/spm/runs/openlane_test/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/spm/runs/openlane_test/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/spm/runs/openlane_test/reports/metrics.csv'.
[WARNING]: There are max fanout violations in the design at the typical corner. Please refer to 'designs/spm/runs/openlane_test/reports/signoff/31-rcx_sta.c
hecks.rpt'.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: 1 warnings found by linter
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd__tapvpwrvgnd_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[WARNING]: There are max fanout violations in the design at the typical corner. Please refer to 'designs/spm/runs/openlane_test/reports/signoff/31-rcx_sta.c
hecks.rpt'.

Basic test passed
gyana@GYANARANJAN:~/OpenLane$ |
```

Figure 2: OpenLane terminal output showing zero DRC violations and flow completion

Physical Sign-Off & Results

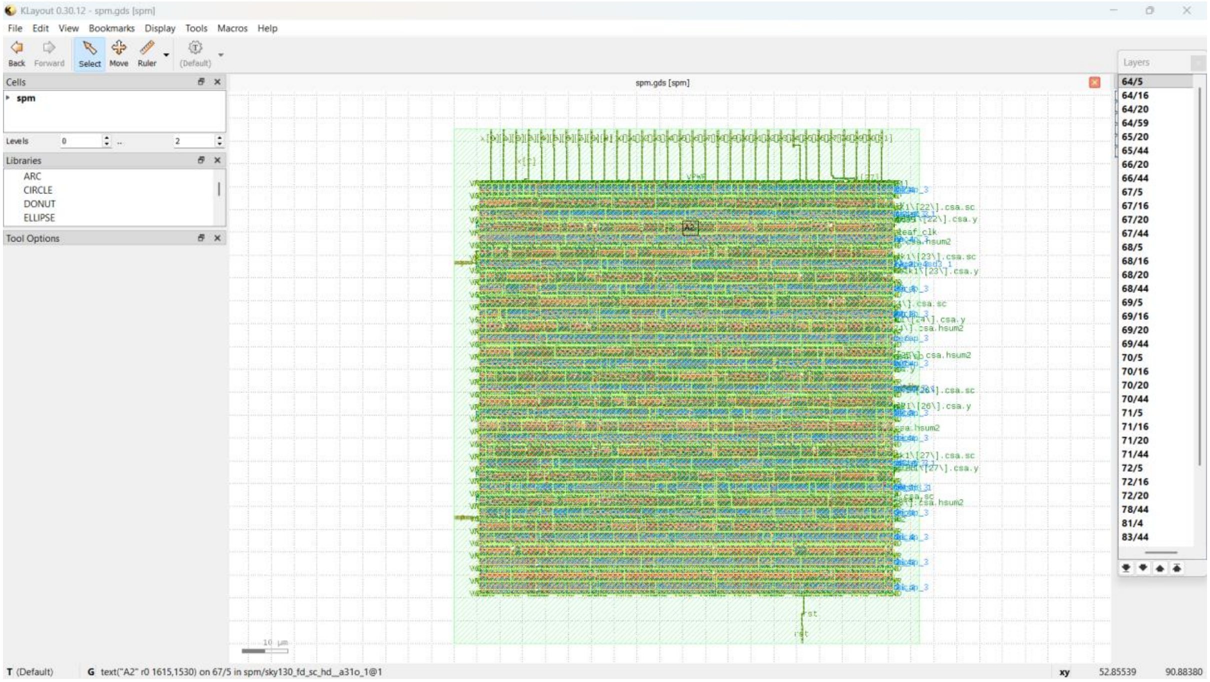
The final design generated clean verification sign-off reports:

- Design Rule Checking (DRC): Magic DRC reported No DRC violations after GDS streaming out.
- Static Timing Analysis (STA): Zero setup violations and zero hold violations confirmed at typical corner conditions.
- GDSII Stream Generation: Successfully streamed out spm.gds mask layout.

Metric	Result	Status
Technology Node	SkyWater 130nm (sky130A)	Active
Setup Violations	0	Passed
Hold Violations	0	Passed
Magic DRC Violations	0	Passed
Final Deliverable	spm.gds	Generated

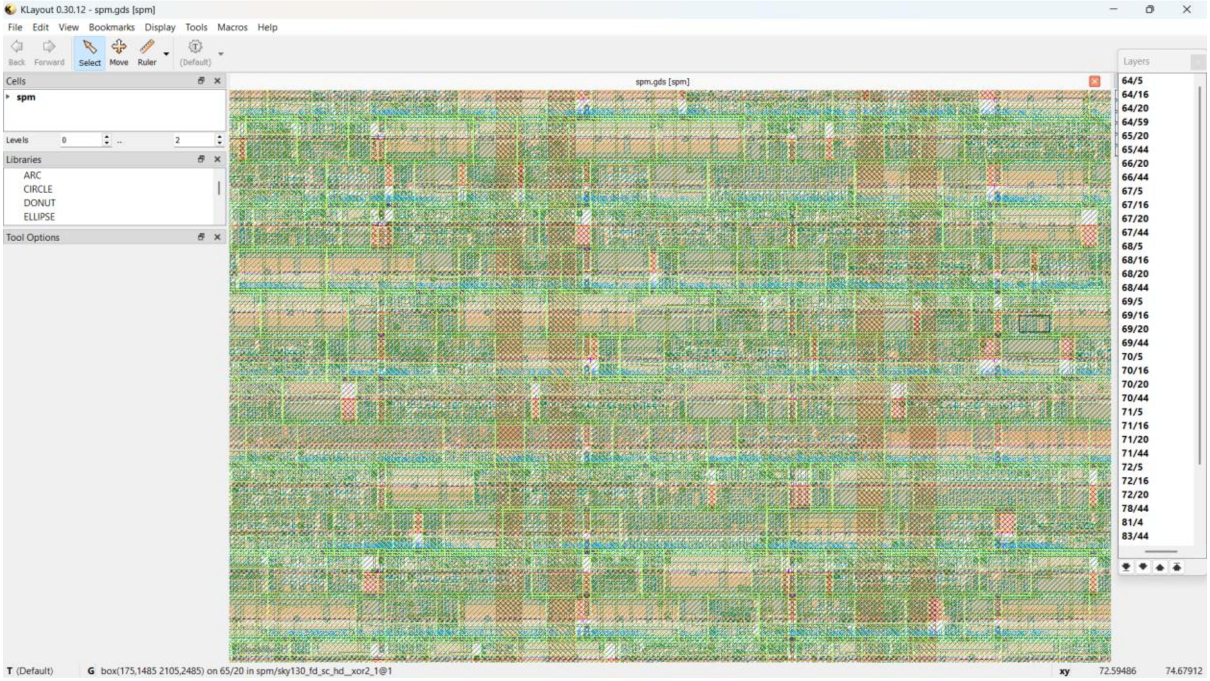
Layout Inspection in KLayout

The generated GDSII stream was visually inspected using KLayout to verify physical geometry, boundary pins, standard cell placement density, and multi-layer interconnects.


KLayout 0.30.12 - spm.gds [spm]
File Edit View Bookmarks Display Tools Macros Help
Back Forward Select Move Ruler (Default)
Cells
spm
Levels 0 .. 2
Libraries
ARC
CIRCLE
DONUT
ELLIPSE
Tool Options
spm.gds [spm]
Layers
64/5 64/16 64/20 64/59 65/20 65/44 66/20 66/44 67/5 67/16 67/20 67/44 68/5 68/16 68/20 68/44 69/5 69/16 69/20 69/44 70/5 70/16 70/20 70/44 71/5 71/16 71/20 71/44 72/5 72/16 72/20 78/44 81/4 83/44
rst
10 µm
T (Default)
xy 52.85539 90.88380

G text("A2" r0 1615,1530) on 67/5 in spm/sky130_fd_sc_hd__a31o_1@1

Figure 3: Top-level view of the SPM core boundary and metal routing in KLayout


KLayout 0.30.12 - spm.gds [spm]
File Edit View Bookmarks Display Tools Macros Help
Back Forward Select Move Ruler (Default)
Cells
spm
Levels 0 .. 2
Libraries
ARC
CIRCLE
DONUT
ELLIPSE
Tool Options
spm.gds [spm]
Layers
64/5
64/16
64/20
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65/20
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70/20
70/44
71/5
71/16
71/20
71/44
72/5
72/16
72/20
78/44
81/4
83/44
T (Default)
xy 72.59486 74.67912

```
G box(175,1485 2105,2485) on 65/20 in spm/sky130_fd_sc_hd__xor2_1@1
```

Figure 4: Zoomed view displaying standard cell instances, power rails, and detailed signal routing

Conclusion

The end-to-end RTL-to-GDSII ASIC design flow for the Serial-Parallel Multiplier was completed successfully using open-source EDA tools on Windows 11 via WSL2. The final design successfully met all electrical, physical, and manufacturability constraints, resulting in a tapeout-ready GDSII mask blueprint.